CASE STUDY

Case study 2

During harvest festival season, tomatoes are available in abundance. How can we make use of this?

Salt:

Salt is employed to control microbial population in foods such as butter, cheese, cabbage, olives, cucumbers, meats, fish and bread. There are four methods of salt curing; dry salting (fish), brimming (vadu manga), low salt fermentation (chilli pickle, sauerkraut from cabbage) and pickling (lime pickle). Sodium chloride or common salt is used primarily as a preservative and flavouring agent.

Vinegar:

Vinegar is a natural preservative. Vinegar is made from a two step process. The first process involves the carbohydrate being converted into alcohol by fermentation. The second step is its conversion to an acetic acid. The acetic acid in vinegar kills microbes and stops food spoilage. Pickling is a common method of using vinegar as a preservative. It is also used to improve the flavour of foods.

Spices and condiments:

These have *bacteriostatic* effect (slowing the growth and multiplication of microbes). The essential oil of spices is inhibitor of microorganism. The inhibitory effects of the spices differ with the kind of spice and the microorganisms being tested. Mustard flour and the volatile oil of mustard, for example, are very effective against *Saccharomyces cerevisiae*. In pickles like avakai and chilli pickle, mustard flour helps in the prevention of the growth of spoilage organisms in the food.

Turmeric powder, tamarind, chilli powder, asafoetida, fenugreek seed, cinnamon and cloves are usually *bacteriostatic*. Ground pepper corn and all spices are less inhibitory than cinnamon and cloves. Extracts of these plants have been shown to be inhibitory to *Bacillus subtilis* and *E. coli*. Allicin is the active principle in onions and garlic that kills bacteria and acts against fungi.

Bacteriostatic effect – prevents the growth of bacteria (i.e., it keeps them in the stationary phase of growth)
Bactericidal effect – means that it kills bacteria (cidal-killing)

Oil:

In addition to salt and several spices, oils are used in making pickles. Spice mixtures and oil are added to the fruit or vegetable. It is allowed to ferment for a month or so. The fermentation process renders fruits soft and the fruit take on the additional aroma and flavour of the spices. Aerobic bacteria and mould growth are prevented by covering the top with oil. Properly prepared and stored pickles can last upto a year or more without spoilage.




Spoiled pickle

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